MIL-F-14888A(AR)
4 December 1981
SUPERSEDING
MIL-F-14888 (MU)
3 March 1969

MILITARY SPECIFICATION

FIN ASSEMBLY, M170

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies for the Department of Defense.

1. SCOPE

1.1 This specification covers Fin Assembly, M170, for use on 81MM Mortar Ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-H-6088	- Heat Treatment of Aluminum Alloys
MIL-A-48078	- Ammunition Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

10551892 - Fin Assembly, M170

(Copies of specifications, standards, drawings and publications listed above or in the item detail specification required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standard Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

ASTM Method E-8 - Tension Testing of Metallic Materials
ASTM Method E-340 - Macroetching Metals and Alloys

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19105).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies).

3. REQUIREMENTS

3.1 General. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. - The components and assemblies shall comply with all requirements specified on drawing (dwg.) 10551892 and associated drawings, and with all requirements specified in applicable specifications and standards.

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3.3 Heat treatment. - Heat treatment equipment, procedures and controls shall be in accordance with MIL-H-6088. The cartridge housing material (forging or bar stock) shall be solution heat treated in such a manner as to avoid overheating of the material or cracking during quenching.

3.4 Hydrostatic test. - The fin housing shall be able to withstand a hydrostatic load of 17,250 ± 250 p.s.i. when tested in accordance with 4.5.4. Each fin housing that successfully meets the requirements of this test shall be permanently stamped with the letter "H" .125 inch maximum height by .01 inch maximum depth on the outside taper surface 3.75 inches from the front face.

3.5 Data requirements. - The contractor shall generate data in accordance with the requirements of the data item descriptions cited herein and in 4.5.3.3, and 4.5.4. Ammunition data cards are required and shall be supplied in accordance with MIL-STD-1167 (see section 6.8). All drawings and revisions of components and assemblies shall be listed on the data card.

3.6 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship

3.7.1 Metal characteristics. - All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, pipe type defects, planar type defects, inclusions, porosity, or any similar characteristics.

3.7.2 Threads. - Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings.

3.7.3 Burrs. - No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.7.4 Foreign matter. - All parts and assemblies shall be free of chips, dirt, grease and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. - The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part description	Drawings	Quantity
Fin Assembly, M170	10551892	10
Fin	10520199	5
Housing	10551922	5
Housing, prior to drilling of flash holes	10551922	10
Increment holder	7549026	5

Following nominal cropping, one specimen suitable for macroetch testing shall be submitted from the rear of both housing and fin as-extruded stock. In addition, 3 tensile specimens shall be submitted from fully heat treated material. Two tensile specimens shall be obtained from extruded or forged housing material stock (as applicable), and one specimen shall be obtained from fin stock. See Table 1, Note d. Microstructure specimens shall be taken from housing material stock as described in 4.5.3.2.

4.3.2 Inspections to be Performed - See MIL-A-48078 and Table 1 specified herein.

4.3.3 Rejection - See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-]4888A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fin Assembly, M170 (dwg. 10551892)				
	Examination for defects	10		3.2	4.4.2.7
	Fin (dwg. 10520199)				
	Examination for defects	5		3.2	4.4.2.5
	Housing (dwg. 10551922)				
	Examination for defects	5		3.2	4.4.2.4
	Housing (prior to drilling flash holes) (dwg. 10551922)				
	Hydrotest	10		3.4	4.5.4
	Hydrotest, to destruction	10 (a)		3.2	4.5.4
	Extrusion Specimens (pre-heat treat)				
	Macroetch examination	2 (b)		3.7.1	4.5.2
	Physical Properties Specimens (housing)				
	Tension test	2		3.1	4.3.1/4.5.3.1.1
	Microstructure examination	2		3.3	4.5.3.2
	Physical Properties (fin)				
	Tension test	1 (c)		3.1	4.3.1/4.5.3.1.2

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-14888A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Holder Increment (dwg. 7439026 Covering Details of Part Number 7549027 Examination for defects	5		3.2	4.4.2.6

NOTES: (a) Test for information only. The same samples shall be used for both hydrostatic tests. (b) One specimen each from the rear of both fin and housing stock. (c) Specimen shall be of sufficient length to allos a standard flat full or sub-size specimen to be cut from stock.

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4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and the following as applicable:

4.4.1.1 Cartridge housing lot. - A cartridge housing lot shall consist of material of the same alloy that has been heat treated in the same heat treating equipment at the same temperature in one unchanged process. The maximum number of pounds in a cartridge housing lot shall consist of 15,000 or fraction thereof. A cartridge housing lot may consist of more than one solution heat treatment batch. A solution heat treat batch shall consist of not more than 1000 pounds of the same alloy that has been thermally treated at the same time by the same method in the same furnace equipment at the same time and temperature in one unchanged process. Tensile and microstructure tests hereinafter prescribed must be selected from and represent each solution heat treatment batch. The identification and segregation of each solution heat treatment batch is required until compliance with the tensile and microstructure requirements is confiremd; at that time the above mentioned segregation is no longer necessary.

4.4.1.2 Fin lot. - The fin lot shall consist of not more than 1000 pounds of extruded stock of the same alloy that has been thermally treated by the same method in the same equipment at the same temperature and time in one unchanged process regardless of the solution heat treating method utilized. Each lot representing material that has been solution heat treated after extrusion, must be solution heat treated at the same time in the same furnace using the same temperature and time in that furnace. The lot thus constituted shall be the basis for selecting the sample specimens for the tension test hereinafter prescribed.

4.4.2. Examination - See MIL-A-48078.

a. Sampling plans. - Sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Tool control dimensions. - Dimensions marked tool control shall be inspected as follows: select and gage

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one fin and one housing from each machine in operation every two hours or fraction thereof and whenever tooling is changed. Prior to any tooling change, the last housing or fin produced from the old tooling shall be gaged for the requirements of the applicable drawing. All material produced prior to periodic inspection shall be rejected and screened whenever a defective tool control dimension is found and corrective action must be taken.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Fin and Housing Extrusion Bars				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special a	Macroetch examination	(a)	100%	3.7.1	4.5.2
Major	None defined				
Minor	None defined				

NOTES:

(a) See paragraph 4.5.2 for sampling procedures. The test shall be performed on each extrusion bar.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-14888A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Fin and Housing Heat Treatment Specimens				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a	Tension test (housing)	2(a)	100%	3.1	4.5.3.1
b	Tension test (fin)	1(a)	100%	3.1	4.5.3.1
c	Microstructure test	2(b)	100%	3.3	4.5.3.2
Major	None defined				
Minor	None defined				

NOTES: (a) Failure of any sample shall cause the respective lot to be rejected.
(b) The tension test samples from the housing samples only shall also be used for this test. Failure of any sample shall be cause for rejection of the representative solution heat treatment batch. See 4.4.1.1.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-14888A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10551922
4.4.2.3	Housing, Prior to Drilling Flash Holes				NEXT HIGHER ASSEMBLY 10551892
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Hydrotest		100%	3.4	4.5.4
Major	None defined				
Minor	None defined				

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-14888A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 10551922
4.4.2.4	Housing, Cartridge				NEXT HIGHER ASSEMBLY 10551892
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Flash hole(s) missing (24)		100%	3.2	4.5.5.1
2.	Hydrostatic test stamp missing		100%	3.4	Visual
Special					
a.	Body thread minor dia. oversize		100%	3.2	Plug gage
b.	Presence of burr or foreign matter in ignition cartridge cavity or any flash hole		100%	3.7	Visual
c.	Metal defective		100%	3.7.1	Visual
d.	Fin thread major diameter undersize		100%	3.2	Snap gage
Major					
101	Runout at front of taper with pitch dia. of body thread and front face		0.40%	3.2	Gage
102	Runout at rear of taper with pitch dia. of body thread and front face		0.40%	3.2	Gage
103	True position of ignition cartridge cavity with pitch dia. of body thread		0.40%	3.2	Gage
104	True position of primer flash hole with ignition cartridge cavity		0.40%	3.2	Gage
105	Pitch diameter of body thread		0.40%	3.2	Gage
106	Pitch diameter of fin thread		0.40%	3.2	Gage
107	Diameter of ignition cartridge cavity		0.40%	3.2	Gage
108	Diameter of primer flash hole		0.40%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-14888A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 10551922
4.4.2.4	Housing, Cartridge				NEXT HIGHER ASSEMBLY 10551892
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109	Chamfer of primer flash hole (both ends) missing or improper		0.40%	3.2	Visual
110	Depth to bottom of body thread cavity		0.40%	3.2	Gage
111	Length of effective body thread, min.		0.40%	3.2	Gage
112	Length of effective fin thread, min.		0.40%	3.2	Gage
113	Thread(s) damaged		0.40%	3.7.2	Visual
114	Flash holes excessive in number		0.40%	3.2	4.5.5.2
115	Depth to bottom of cartridge cavity		0.40%	3.2	Gage
Minor					
201	Diameter at forward basic length		0.65%	3.2	Gage
202	Diameter at rear basic length		0.65%	3.2	Gage
203	Diameter of body thread counterbore		0.65%	3.2	Gage
204	Depth of body thread counterbore		0.65%	3.2	Gage
205	Length from rear face to fin seat		0.65%	3.2	Gage
206	Length, total		0.65%	3.2	Gage
207	Radii or chamfer missing or incorrect		0.65%	3.2	Visual (See 6.6)
208	Surface finish improper		0.65%	3.2	Visual (See 6.7)
209	Presence of burrs		0.65%	3.2	Visual
210	Foreign matter		0.65%	3.2	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 10520199
4.4.2.5	Fin				NEXT HIGHER ASSEMBLY 10551892
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Metal defective		100%	3.1	Visual
Special					
a.	Minor dia. of housing thread oversize		100%	3.2	Plug gage
Major					
101	True position of primer thread counterbore with pitch dia. of primer thread		0.40%	3.2	Gage
102	Straightness of fin blades including twist, after canting blades		0.40%	3.2	Gage
103	Pitch dia. of housing thread		0.40%	3.2	Gage
104	Pitch dia. of primer thread		0.40%	3.2	Gage
105	Minor diameter of primer thread		0.40%	3.2	Gage
106	Pitch dia. of through thread (if applicable)		0.40%	3.2	Gage
107	Minor dia. of through thread (if applicable)		0.40%	3.2	Gage
108	Dia. of housing thread counterbore		0.40%	3.2	Gage
109	Large diameter of fin blades		0.40%	3.2	Gage
110	Small diameter of fin blades		0.40%	3.2	Gage

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PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 10520199
4.4.2.5	Fin				NEXT HIGHER ASSEMBLY 10551892
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111	Diameter of primer thread counter-bore (large)		0.40%	3.2	Gage
112	Diameter of primer thread counter-bore (small)		0.40%	3.2	Gage
113	Depth of housing thread counterbore		0.40%	3.2	Gage
114	Depth of primer thread counterbore(large)		0.40%	3.2	Gage
115	Length of effective housing thread, min.		0.40%	3.2	Gage
116	Length of effective primer thread, min.		0.40%	3.2	Gage
117	Length from front face to rear face of hub		0.40%	3.2	Gage
118	Fin blades canted in improper direction		0.40%	3.2	Visual
119	Tangency of radius with hub diameter and fin blade surface		0.40%	3.2	Visual (See 6.6)
120	Thread(s) damaged		0.40%	3.7.2	Visual
Minor					
201	Angle at front of fin blades		0.65%	3.2	Gage
202	Angle between fin blades		0.65%	3.2	Gage
203	Diameter of hub		0.65%	3.2	Gage
204	Dia. of inside between thread (if applicable)		0.65%	3.2	Gage
205	Length of cant		0.65%	3.2	Gage
206	Length, total		0.65%	3.2	Gage
207	Thickness of fin blades		0.65%	3.2	Gage

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PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 10520199
4.4.2.5	Fin				NEXT HIGHER ASSEMBLY 10551892
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
208	Radii or chamfers missing or incorrect		0.65%	3.2	Visual (See 6.6)
209	Surface finish improper		0.65%	3.2	Visual (See 6.7)
210	Presence of burrs		0.65%	3.7.3	Visual
211	Foreign matter		0.65%	3.7.4	Visual

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MIL-F-14888A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Holder, Increment		7549026
			NEXT HIGHER ASSEMBLY
			10551892

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Diameter, outside, major		0.65%	3.2	Gage
202	Diameter, outside, hub		0.65%	3.2	Gage
203	Diameter, inside		0.65%	3.2	Gage
204	Thickness		0.65%	3.2	Gage
205	Length of eye(s)		0.65%	3.2	Visual (See 6.6)
206	Width of eye(s)		0.65%	3.2	Visual (See 6.6)
207	Width of eye neck		0.65%	3.2	Visual (See 6.6)
208	Radii missing or incorrect		0.65%	3.2	Visual (See 6.6)
209	Surface finish improper		0.65%	3.2	Visual (See 6.6)
210	Eye(s) missing		0.65%	3.2	Visual
211	Metal defective		0.65%	3.1	Visual
212	Protective coating inadequate or damaged		0.65%	3.2	Visual
213	Presence of burrs		0.65%	3.7.3	Visul
214	Foreign matter		0.65%	3.7.4	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.7	Fin Assembly		1 OF 1	10551892
				NEXT HIGHER ASSEMBLY --

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Primer flash hole missing or obstructed		100%	3.2	Visual
2.	Metal defective		100%	3.1	Visual
Major					
101	True position of fin blades with pitch dia. of body thread and front face		0.40%	3.2	Gage
102	True position of primer thread minor dia. with pitch dia. of body thread and front face (if applicable)		0.40%	3.2	Gage
103	Length, total, max.		0.40%	3.2	Gage
104	Fin blades distorted or damaged		0.40%	3.2	Visual (See 6.6)
105	Increment holder missing or damaged		0.40%	3.2	Visual
106	Thread(s) damaged		0.40%	3.7.2	Visual
107	Torque test		0.40%	3.2	4.5.6
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.2	Visual
202	Presence of burrs		0.65%	3.7.3	Visual
203	Foreign matter		0.65%	3.7.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-14888A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Packed Carton Prior to Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Support(s) missing or damaged		0.65%	3.2	Visual
202	Divider(s) missing or damaged		0.65%	3.2	Visual
203	Improperly packed		0.65%	3.2	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Packed Carton After Sealing		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Cartons damaged (cut or damaged through all layers)		0.65%	3.2	Visual
203	Carton not properly sealed		0.65%	3.2	Visual
204	Marking missing, incorrect or illegible		0.65%	3.2	Visual

NOTES:

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4.4.3 Testing. - Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed in this specification is identified either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the Quality Conformance Inspection tables herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Materials, components and processes. - Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. - The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Macroetch examination. - A full section macroetch specimen shall be cut from the back end of the bar representing the last extruded material from each individual extrusion billet. Failure to comply with the requirements of 3.7.1 will require an additional length of discard from the defective extrusion end and testing of an additional macroetch specimen taken from the end of the remaining length of extruded material. This procedure must be continued until a sufficient length of defective extrusion has been discarded and the requirements of 3.7.1 are met. Suitable identification of each length of extrusion and corresponding macroetch must be maintained until the extruded material is accepted. The specimen shall be macroetched using the criteria prescribed in ASTM Method E-340 except that sodium hydroxide shall be used as the macroetching reagent.

4.5.3 Mechanical properties

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4.5.3.1 Tension test

4.5.3.1.1 Cartridge housing lot. - Two specimens, one from the hottest and one from the coldest zone of each solution heat treatment batch shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Specimens taken shall conform to the largest obtainable standard round type specimen as prescribed by ASTM Method E-8 and approved by the procuring agency; the Government shall not be restricted from testing additional specimens to determine that the mechanical properties conform to drawing requirements. The criteria for performing this test shall be as prescribed by ASTM Method E-8.

4.5.3.1.2 Fin stock extrusion lots. - From each fin lot (as defined by 4.4.1.2), at least one specimen shall be subjected to testing after all heat treatment has been completed. Failure of any tension test specimen to comply with the yield strength or elongation requirement of the applicable drawing shall be cause for rejection of the lot. The test criteria shall be as prescribed by ASTM Method E-8 and each test specimen shall be of sufficient length to allow a standard flat full or sub-size specimen to be cut from stock or a fin blade. The Government shall not be restricted from testing additional specimens to determine that the mechanical properties meet the drawing requirements for yield strength and elongation.

4.5.3.2 Microstructure examination. - From each of the housing specimens submitted for tension testing in 4.5.3.1.1 and 4.3.1, one wafer, approximately one-half inch in length shall be cut longitudinally and prepared for metallographic examination. When so prepared, the specimen shall be etched with Keller's reagent and examined by scanning under a microscope at a minimum magnification of 500 diameters. Overheating shall be evidenced by the presence of rosettes in the microstructure or incipient melting at the grain boundaries (see 6.5). Evidence of the above shall cause rejection of the representative solution heat treatment batch.

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4.5.3.3 Quality inspection report. - The contractor shall submit a quality inspection test, demonstration, and evaluation report in accordance with Data Item Description DI-R-1724 (tailored) giving the detailed test results from the above tests (4.5.3.1 and 4.5.3.2).

4.5.4 Hydrostatic test. - After final machining but prior to drilling flash holes and prior to gaging, each cartridge housing shall be subjected to a hydrostatic test to determine compliance with the requirements of 3.4 and 3.7.1. Failure to meet the requirements of 3.4 and 3.7.1 shall be cause for rejection of the housing. The housing shall be tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached and held for three seconds. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours of continuous operation. It shall also be calibrated prior to restarting the operation after a lay-off of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the pressure was actually lower than required, the housings tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. Rupture, cracking, or bulging shall be evidence of failure and cause for rejection of the housing. Each housing which has successfully passed the test shall be stamped in accordance with 3.4. For first article inspection only, the test pressure shall be increased until the housing ruptures and the maximum pressure shall be recorded for information purposes. The contractor shall submit a monthly quality defect report in accordance with Data Item Description DI-R-1721 summarizing the defects found.

4.5.5 Flash hole inspection

4.5.5.1 Missing flash holes. - Each fin housing shall be inspected by an electronic and/or mechanical system capable of determining the presence of the required number of flash holes (see drawing).

4.5.5.2 Excess flash holes. - Each fin housing shall be inspected by a method capable of determining the presence of flash holes in excess of the drawing requirement.

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